Overview

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3 Simple Random Sampling

- Simple random sampling (SRS) is the **gold standard** of probability sampling designs.
- Let U be a population of N elements. The sample size n $(0 < n < N)$ under SRS can be expressed by the **sampling fraction**,

$$f = \frac{n}{N}.$$

- The **sampling design** p_{SRS} was introduced in Chapter 2 and is given, for $s \in \mathscr{S}$, by

$$p_{\mathrm{SRS}}(s) = \begin{cases} 1/\binom{N}{n} & \text{if } |s| = n, \\ 0 & \text{otherwise.} \end{cases}$$

- The (first-order sample) **inclusion probabilities** under p_{SRS} have the following form (see Chapter 2):

$$\pi_i = \mathbb{P}(i \in s) = \frac{n}{N} \qquad \text{for every } i \in U.$$

- **Selecting** an SRS in R:

```
sample(x, size, replace = FALSE, prob = NULL)
```

- For example,

```
> set.seed(2048)
> sample(1:100, 10)
[1] 23 97 88 96 43 27 57 67 31 12
```

3.1 Estimation

- Let s be an SRS of size n from U. The variable of interest is denoted by y_i.
- An *unbiased* estimator of the **population mean** $\bar{y}_U$ is

$$\bar{y} = \frac{1}{n} \sum_{i \in s} y_i.$$

Aside (Unbiasedness).

To show that $\bar{y}$ is an unbiased estimator of $\bar{y}_U$, we express the estimator as a sum over $i \in U$,

$$\bar{y} = \frac{1}{n} \sum_{i \in s} y_i = \frac{1}{n} \sum_{i \in U} \mathbb{1}\{i \in s\} y_i,$$

Aside (Cont.).

where $\mathbb{1}\{i \in s\}$ is the sample inclusion indicator. Under the sampling design, $\mathbb{1}\{i \in s\}$ is a Bernoulli random variable. Under p_{SRS}, its expected value is

$$\mathbb{E}_p\big(\mathbb{1}\{i \in s\}\big) = \pi_i = \frac{n}{N}$$

for every $i \in U$. Hence,

$$\mathbb{E}_p(\bar{y}) = \frac{1}{n} \sum_{i \in U} \mathbb{E}_p\big(\mathbb{1}\{i \in s\}\big)\, y_i = \frac{1}{n} \sum_{i \in U} \pi_i\, y_i = \frac{1}{n} \sum_{i \in U} \frac{n}{N}\, y_i = \bar{y}_U,$$

as required.

- The **theoretical variance** of the estimator $\bar{y}$ (var or var_p if we want to highlight the dependence on the design p) is

$$\text{var}(\bar{y}) = \left(1 - \frac{n}{N}\right) \frac{\sigma_y^2}{n} = (1-f)\frac{\sigma_y^2}{n},$$

where σ_y^2 is the population variance of the y-variable,

$$\sigma_y^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \bar{y}_U)^2.$$

- The factor $(1-f)$ is called the **finite population correction** (FPC). Without this factor, $\text{var}(\bar{y})$ would equal the familiar variance formula for the sample mean from classical statistics.

- In the **special case** $n = N$ (a census), $\text{var}(\bar{y}) = 0$.

- An unbiased **variance estimator** for $\bar{y}$ that requires only the observed values y_i, $i \in s$, is

$$\widehat{\mathsf{var}}(\bar{y}) = \left(1 - \frac{n}{N}\right) \frac{\widehat{\sigma}_y^2}{n},$$

where

$$\widehat{\sigma}_y^2 = \frac{1}{n-1} \sum_{i \in s} (y_i - \bar{y})^2.$$

- The estimated **standard error** $\widehat{\mathsf{SE}}(\bar{y})$ is defined as the square root of $\widehat{\mathsf{var}}(\bar{y})$.

- The estimated **coefficient of variation** is, for $\bar{y} > 0$,

$$\widehat{\mathsf{CV}}(\bar{y}) = \frac{\widehat{\mathsf{SE}}(\bar{y})}{\bar{y}}.$$

Population total

- The **population total** t_y can be estimated unbiasedly by

$$\widehat{t}_y = N\,\bar{y}.$$

Proportion

- Let $b_i = \mathbb{1}\{i \in B\}$ be the indicator variable for the **dichotomous characteristic** B, $i \in U$ (e.g., whether a person is employed, "yes" $= 1$ and "no" $= 0$).
- The **population proportion** P_b is

$$P_b = \frac{1}{N}\sum_{i \in U} b_i = \frac{\text{number of units with characteristic } B}{N}.$$

3.2 Statistical inference

- Recall the **central limit theorem** from Chapter 2. Under suitable regularity conditions,

$$\frac{\bar{y} - \bar{y}_U}{\widehat{\mathsf{SE}}(\bar{y})}$$

  is approximately $N(0,1)$-distributed for sufficiently large n.

- Thus, for sufficiently large n, an approximate $100(1-\alpha)\%$ **confidence interval** (CI) for $\bar{y}_U$ is

$$\bar{y} \pm z_{1-\alpha/2}\widehat{\mathsf{SE}}(\bar{y}), \tag{3.1}$$

  where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the $N(0,1)$ distribution.

- The confidence interval is based on a **normal approximation**.
  - There is no *universal* minimum sample size for this approximation to be accurate. Its accuracy depends on the finite population, in particular on skewness and extreme values.
  - Because such features are *common* in survey applications, simple rules such as $n \geq 30$" or $n \geq 50$" should be used with caution.
- A **practical question** is therefore how large n should be for the normal approximation to be sufficiently accurate.
- For populations with marked **positive skewness**, Cochran (1977, p. 42) proposed the **crude rule**

$$n > 25, G_1^2,$$

where G_1 denotes the population skewness.

- Population skewness is defined as

$$G_1 = \frac{1}{N} \sum_{i \in U} \left(\frac{y_i - \bar{y}_U}{\sigma_y} \right)^3$$

- Cochran emphasized that this is only a **crude rule of thumb**, not a general criterion for the normal approximation.

- Sugden et al. (2000) proposed the more conservative rule

$$n_{\min} = 28 + 25\, G_1^2.$$

Confidence interval for an estimated proportion

- To construct a $100(1-\alpha)\%$ CI for the estimated proportion $\widehat{P}$, we can replace $\bar{y}$ by $\widehat{P}$ in (3.1), giving

$$\widehat{P} \pm z_{1-\alpha/2} \sqrt{(1-f)\,\frac{\widehat{P}(1-\widehat{P})}{n-1}}.$$

- The resulting CI is called the **Wald interval**. It can be used under the usual conditions, i.e., when $nP \geq 5$ and $n(1-P) \geq 5$.

- The Wald interval can behave erratically and is considered unreliable when n is small or P is close to 0 or 1. Brown et al. (2001) therefore recommend the **Wilson interval**; it is implemented in the function `svyciprop()` of the R package `survey`.

3.3 Sample size determination

- One of the first questions in survey planning is: **How large should the sample be?**
- The answer depends mainly on three **factors**:
  - the total survey budget;
  - the cost of observing one unit;
  - the required precision of the estimates.
- In this section, we consider only **sample size determination** based on the **required precision**.

- Even when a survey collects **many variables**, it is often possible to focus on one or two variables of **primary interest** and use them to determine the sample size.
- In what follows, we assume that the required precision is specified for the estimated **population mean** $\bar{y}_U$.
- Let $\epsilon > 0$ denote the **margin of error**. The precision requirement can then be stated as follows: **Choose the sample size such that**

$$\mathbb{P}\left(|\bar{y} - \bar{y}_U| > \epsilon\right) \leq \alpha, \tag{3.2}$$

  where the probability α (e.g., 5%) and $\epsilon > 0$ are specified in advance.
- By the **central limit theorem** (see Theorem 1 in Chapter 2), $(\bar{y} - \bar{y}_U) / \widehat{\mathrm{SE}}(\bar{y})$ has an approximately $N(0,1)$ distribution for a sufficiently large sample size.

- Using the $N(0,1)$ approximation, we can rewrite (3.2) and obtain the **initial estimate** that the sample size must be at least

$$n_0 = \left(\frac{z_{1-\frac{\alpha}{2}} \cdot \sigma_y}{\epsilon} \right)^2, \tag{3.3}$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the $N(0,1)$ distribution and σ_y is the standard deviation of the y-variable.

- **Alternative derivation**: Set the half-width of the $100(1-\alpha)\%$ confidence interval for $\bar{y}$ equal to ϵ:

$$\bar{y} \pm \underbrace{z_{1-\alpha/2}\, \widehat{\mathsf{SE}}(\bar{y})}_{=\epsilon}.$$

- In practice, n_0 is **rounded up** to the next integer.

- If $n_0 > N$, we set $n_0 = N$ (a census).

- The estimate in (3.3) is suitable when the resulting sampling fraction n_0/N is negligible (e.g., $n_0/N < 0.01$). Otherwise, we use the **refined estimate**

$$n = \frac{n_0}{1 + n_0/N}, \tag{3.4}$$

  which accounts for the **finite population correction**.

- If the population is very large, typically $n \ll N$.

  $\Rightarrow$ In this case, the required sample size is determined by the **absolute size** of n, not by the ratio n/N.

- In practice, the (population) standard deviation σ_y in (3.3) is **unknown** and must be estimated. The following options are available:

  1. conduct a **pilot study**;
  2. use information from **previous studies** or the literature;
  3. use a **rough estimate** if the first two options are not feasible.

Anticipated nonresponse

- **Nonresponse and refusals** are common in survey practice.
- The **expected amount** of nonresponse must therefore be taken into account at the planning stage.
- Let r denote the anticipated response rate, $r \in (0, 1]$. The **gross sample size** n^* is

$$n^* = \frac{n}{r},$$

  where n is the **net sample size** determined using the formulas above.
- In practice, the response rate r can usually only be estimated **roughly** from previous experience.

3.4 Estimation for subpopulations – Unplanned domains

- Let $U_d \subset U$ be a **subpopulation** of U that is of particular interest for the analysis (a domain of interest).
- We also refer to U_d simply as a **domain**.

Language region	Number	Proportion
German	1 260	66%
French	572	30%
Italian	70	4%
Total	1 902	100%

Example. Population of ophthalmologists in Switzerland with a license to practice, by language region (MedReg, 2022)

- Assume that domain U_d is **unplanned**. This means that
  - the elements of U_d **cannot be identified** in advance, and
  - its population size N_d is **unknown** ($P_d = N_d/N$ is unknown).
- We want to estimate the **mean** $\bar{y}_{U_d}$ in the **subpopulation** U_d.
- First, note that $\bar{y}_{U_d}$ can be written as

$$\bar{y}_{U_d} = \frac{\sum_{i \in U} \mathbb{1}_{id}\, y_i}{\sum_{i \in U} \mathbb{1}_{id}},$$

where

$$\mathbb{1}_{id} = \mathbb{1}\{i \in U_d\} = \begin{cases} 1 & \text{if } i \in U_d, \\ 0 & \text{otherwise.} \end{cases}$$

Thus, $\bar{y}_{U_d}$ is the **ratio** of two population totals.

- The **sample size** n_d of the subsample s_d falling in U_d ($s_d = s \cap U_d$) is known **only after** the sample s has been selected.
- Using the indicator variable

$$\mathbb{1}_{id}^s = \mathbb{1}\{i \in s_d\} = \begin{cases} 1 & \text{if } i \in s_d, \\ 0 & \text{otherwise,} \end{cases} \qquad i \in U,$$

the **domain sample size** can be written as

$$n_d = \sum_{i \in U} \mathbb{1}_{id}^s.$$

- Since $\mathbb{1}_{id}^s$ is a function of the random sample s, n_d is a **random variable** with expected value nP_d (hypergeometric distribution).

Example 11 (Ophthalmologists).

Suppose an SRS of size $n = 500$ is drawn from the population of ophthalmologists. The expected realized sample size n_d for Italian-speaking Switzerland is approximately 18. With probability 95%, n_d lies in the interval $[12, 26]$.

- An estimator of $\bar{y}_{U_d}$ is
$$\bar{y}_d = \frac{\sum_{i \in s_d} y_i}{n_d}, \qquad n_d > 0.$$
- This estimator is a **ratio** of two estimated totals; see Chapter 7.
- Although this estimator has the form of a ratio of two estimated totals, it has a useful special property under SRS.
- **Conditional** on $n_d = m > 0$, the domain sample s_d is an **SRS of size m from U_d**.

- Hence,
$$\mathbb{E}_p(\bar{y}_d \mid n_d = m) = \bar{y}_{U_d}.$$

- Thus, $\bar{y}_d$ is unbiased for the domain mean, **conditional** on observing at least one unit from the domain.

- The main difficulty with an **unplanned domain** is that n_d is random.
  - If the domain is small, n_d may be small, resulting in **low precision**.
  - It is even possible that $n_d = 0$, in which case $\bar{y}_d$ cannot be computed.

- If reliable estimates are required for a particular domain, it should, whenever possible, be taken into account when designing the sample (**planned domain**; see Chapter 4).

Appendix: Determining the required sample size (derivation)

- Let $\widehat{P}$ denote the estimator of the population proportion P under the design p_{SRS}. Ignoring the finite population correction, its estimated standard error is

$$\widehat{\mathsf{SE}}(\widehat{P},) = \sqrt{\frac{\widehat{P}(1-\widehat{P})}{n_0 - 1}} \approx \sqrt{\frac{\widehat{P}(1-\widehat{P})}{n_0}}, \tag{C.1}$$

  where the approximation is appropriate when n_0 is sufficiently large.

- The precision requirement can now be stated as follows: **Choose the sample size such that**

$$\mathbb{P}\left(|\widehat{P} - P| > \epsilon\right) \leq \alpha, \tag{C.2}$$

  where the probability α (e.g., 5%) and the **margin of error** $\epsilon > 0$ are specified in advance.

- The requirement in (C.2) is **equivalent** to

$$\mathbb{P}\big(|\widehat{P} - P| \leq \epsilon\big) \geq 1 - \alpha.$$

- For sufficiently large sample sizes, the **central limit theorem** gives

$$Z = \frac{\widehat{P} - P}{\widehat{\mathsf{SE}}(\widehat{P})} \approx N(0,1).$$

- Hence, using the **symmetry** of the standard normal distribution,

$$\frac{\epsilon}{\widehat{\mathsf{SE}}(\widehat{P})} \geq z_{1-\alpha/2}, \tag{C.3}$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the $N(0,1)$ distribution.

- Substituting the approximation from (C.1) into (C.3) gives the initial sample size

$$n_0 = \frac{P_0(1-P_0)\, z_{1-\alpha/2}^2}{\epsilon^2}, \tag{C.4}$$

where P_0 is an **anticipated value** of the population proportion. If the sampling fraction n_0/N is not negligible, the **finite population correction** gives

$$n = \frac{n_0}{1+(n_0-1)/N}. \tag{C.5}$$

- Substituting (C.4) into (C.5) gives

$$n \geq \frac{N P_0(1-P_0) z_{1-\alpha/2}^2}{\epsilon^2(N-1)+P_0(1-P_0) z_{1-\alpha/2}^2}.$$

- The resulting sample size is rounded up to the next integer.

Literature

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